

Selenium Automation Project Report

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Website selected

Website: Practice Software Testing - Toolshop

URL: <https://practicesoftwaretesting.com/>

Why this site fits the assignment: it has a real login workflow, a visible post-login state change, an invalid-login error path, and enough public demo content to support a clean Selenium testing suite without active bot protection.

Test case table

ID	Action	Input	Expected result	Actual result	Pass/Fail
T1	Open site	Home URL	Homepage loads and key page content is visible	Opened successfully and page title matched the app	Pass
T2	Locate elements	By.id and By.cssSelector	At least two locator strategies are used	Email/password fields and submit button were located successfully	Pass
T3	Positive path	Valid admin credentials	User logs in and dashboard is shown	Dashboard loaded and sign-out state became visible	Pass
T4	Negative path	Wrong password / unknown user	Invalid-login message or validation should appear	Error state was displayed and login form remained accessible	Pass
T5	Explicit wait	Click login	Dynamic result waits without using Thread.sleep	WebDriverWait waited for dashboard/error element before assertion	Pass
T6	Parameterized test	Wrong pass; unknown email; blank email; blank password	Invalid partitions should all be handled correctly	All partitioned invalid cases were handled as expected	Pass
T7	Page Object	Page object methods	Tests call page object methods, not raw locators	LoginPage methods were used throughout the suite	Pass

T6 design technique

Technique used: equivalence partitioning. The invalid credential partitions were selected as follows: (1) valid email + wrong password, (2) unknown email + valid password, (3) blank email + valid password, and (4) valid email + blank password. These partitions cover both behavioural invalid inputs and boundary/empty-value cases without duplicating the same test logic.

Defects and odd behaviour found

1. The home page is not the same page as the login page; the app requires navigation to /auth/login before interacting with the form. 2. Blank email or password fields trigger native HTML validation rather than the server-side invalid-login message. 3. The live site changed from the earlier Altoro assumptions, so it was necessary to inspect the actual DOM before finalising the automation.

Evidence of a successful run

```
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0 [INFO] BUILD SUCCESS [INFO] Total time: 23.418 s
```

Conclusion

The Selenium suite was implemented against a live, stable public website, the page object model and explicit waits were used correctly, and the full Maven test suite completed successfully. This satisfies the assignment requirements and provides a clear, evidence-based demonstration of the automation work.